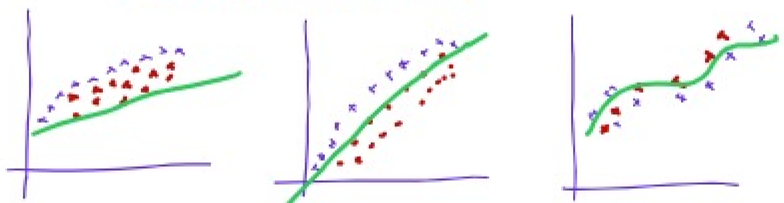
Overfiting and underfiting



Underfiting — Genralize model — Overfiting model

Underfiting	Genralize model	Overfiting model
→ High Bias	→ Low Bias	→ Low Bias
→ Low Variance	→ Low Variance	→ high Variance

Bias : Specific Related to training Data :
Variance : Specific Related to test data :

Overfiting	Under fiting
Training Accuracy = 98%	Traning Acc → 62%
Test Accuracy = 20%	Testing Acc = 60%
Low Bias	high Bias
High variance	high Variance

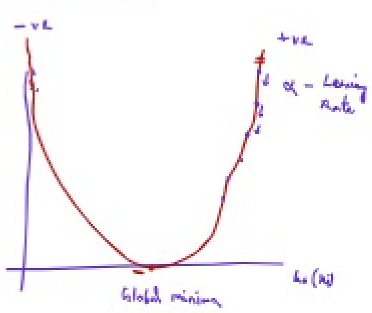
L2 - Regurlization (Ridge & Lasso Regression)

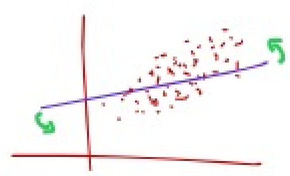
$\int$

Gradient Desent

cost $f^{(n)} = 0$

θ_0 , θ_i





Repeat Conversion theorem

$$\theta_j : = \theta_j - \alpha \frac{\delta}{\delta \theta_j} (J\theta)$$

θ_j = cost function

$\theta_i = \theta_i - \alpha (+ve)$ } Decrease

$\theta_i = \theta_i + \alpha (-ve)$ } Increase

α = Learning Rate

Ridge Regulurazition

$$\text{Cost } f^n = \frac{1}{2m} \sum (h_\theta x_i - y_i)^2 + \underset{\text{Lambda}}{\lambda} (\text{slope})^2$$

← Penalizing Parameter

$= 0 + (2)^2 = 4$ Overfiting ↓↓↓

$= -4 + 4 = 0$

Lasso Regutarazation

$$\text{Cost} = \frac{1}{2m} \sum (h_\theta x_i - y_i)^2 + \lambda \, |\text{slope}|$$ → Underfiting ↓↓↓